Dixell XW60K — Modbus RTU Register Map

Source: XWEB EVO library LIBPackage-20190221-XW60K_002C00E000CA.tar.gz (v2019) | Serial: **9600 8N1** | Protocol: **Modbus RTU** | Polling interval: 900 ms | Values: signed 16-bit integers

Temperature Probes

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC03	0x0100	256	PbReg	Regulation Probe	°C	×0.1	INT16	R	*Display temperature / active regulation probe*
FC03	0x0108	264	Pb1	Probe 1	°C	×0.1	INT16	R	*HACCP probe (main temperature sensor)*
FC03	0x0109	265	Pb2	Probe 2	°C	×0.1	INT16	R	
FC03	0x010A	266	Pb3	Probe 3	°C	×0.1	INT16	R	
FC03	0x010B	267	Pb4	Probe 4	°C	×0.1	INT16	R	

Setpoint

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC03	0x0360	864	SetPoint	Setpoint (read)	°C	×0.1	INT16	**R/W**	*User setpoint; write via FC16*
FC03	0x0600	1536	SetReg	Regulation Setpoint	°C	×0.1	INT16	R	*Active regulation setpoint (may differ from user SP)*

Device Status

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC01	0x0200	512	On	Device On	bool	—	BOOL	R	*1=device powered on*
FC01	0x0201	513	Defrost	Defrost Active	bool	—	BOOL	R	*1=defrost cycle in progress*
FC01	0x0202	514	FastFreezing	Fast Freeze	bool	—	BOOL	R	*1=fast freeze mode active*
FC01	0x0203	515	Keyboard	Keyboard Locked	bool	—	BOOL	R	*1=keypad locked*
FC01	0x0204	516	EnergySaving	Energy Saving	bool	—	BOOL	R	*1=energy saving mode active*
FC01	0x0206	518	Buzzer	Buzzer	bool	—	BOOL	R	*1=buzzer sounding*
FC01	0x0207	519	DigitalInput	Digital Input	bool	—	BOOL	R	*State of digital input*

Alarms

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC01	0x0208	520	ErrorPb1	Probe 1 Error	bool	—	BOOL	R	*Probe 1 fault (short/open)*
FC01	0x0209	521	ErrorPb2	Probe 2 Error	bool	—	BOOL	R	
FC01	0x020A	522	ErrorPb3	Probe 3 Error	bool	—	BOOL	R	
FC01	0x020B	523	ErrorPb4	Probe 4 Error	bool	—	BOOL	R	
FC01	0x020C	524	HTA	High Temp Alarm	bool	—	BOOL	R	*Temperature exceeds high alarm threshold*
FC01	0x020D	525	LTA	Low Temp Alarm	bool	—	BOOL	R	*Temperature below low alarm threshold*
FC01	0x020E	526	CHTA	Condenser High Temp Alarm	bool	—	BOOL	R	*Condenser temperature too high*
FC01	0x020F	527	CLTA	Condenser Low Temp Alarm	bool	—	BOOL	R	
FC01	0x0210	528	ExternalAlarm	External Alarm	bool	—	BOOL	R	*State of external alarm input*
FC01	0x0211	529	SevereAlarm	Severe Alarm	bool	—	BOOL	R	*Critical alarm active*
FC01	0x0212	530	OpenDoor	Door Open	bool	—	BOOL	R	*Door/lid open alarm*
FC01	0x0213	531	EEPROMFailure	EEPROM Failure	bool	—	BOOL	R	*EEPROM read/write error*

Relay Outputs

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC01	0x0216	534	OnOffOUT	On/Off Relay	bool	—	BOOL	R	*Main on/off relay state*
FC01	0x0217	535	DefrostOUT	Defrost Relay	bool	—	BOOL	R	
FC01	0x0218	536	Defrost2OUT	Defrost 2 Relay	bool	—	BOOL	R	
FC01	0x0219	537	Alarm	Alarm Relay	bool	—	BOOL	R	*Alarm relay energized*
FC01	0x021A	538	Light	Light Relay	bool	—	BOOL	R	
FC01	0x021B	539	Fan	Fan Relay	bool	—	BOOL	R	
FC01	0x021C	540	Aux	Aux Relay	bool	—	BOOL	R	
FC01	0x021D	541	DeadBand	Dead Band	bool	—	BOOL	R	*Controller in dead band (neither cooling nor heating)*
FC01	0x021E	542	Compressor1	Compressor 1	bool	—	BOOL	R	*Compressor 1 running*
FC01	0x021F	543	Compressor2	Compressor 2	bool	—	BOOL	R	

Commands

FC	Addr (hex)	Addr (dec)	Name	Label	Unit	Scaling	Type	R/W	Notes
FC05	0x0200	512	DeviceON	Device On	—	0xFF00	COIL	W	*Write 0xFF00 to switch device on*
FC05	0x0200	512	DeviceOFF	Device Off	—	0x0000	COIL	W	*Write 0x0000 to switch device off*
FC05	0x0201	513	ActiveDefrost	Start Defrost	—	0xFF00	COIL	W	*Trigger manual defrost*
FC05	0x0202	514	FastFreezeON	Fast Freeze ON	—	0xFF00	COIL	W	
FC05	0x0202	514	FastFreezeOFF	Fast Freeze OFF	—	0x0000	COIL	W	
FC05	0x0203	515	KeyboardLOCK	Lock Keyboard	—	0xFF00	COIL	W	
FC05	0x0203	515	KeyboardUN-LOCK	Unlock Keyboard	—	0x0000	COIL	W	
FC05	0x0204	516	AlarmMute	Mute Alarm	—	0xFF00	COIL	W	*Silence buzzer*
FC05	0x0206	518	EnergysavingON	Energy Saving ON	—	0xFF00	COIL	W	
FC05	0x0206	518	EnergysavingOFF	Energy Saving OFF	—	0x0000	COIL	W	
FC05	0x021A	538	LightON	Light ON	—	0xFF00	COIL	W	
FC05	0x021A	538	LightOFF	Light OFF	—	0x0000	COIL	W	
FC05	0x021C	540	AuxON	Aux ON	—	0xFF00	COIL	W	
FC05	0x021C	540	AuxOFF	Aux OFF	—	0x0000	COIL	W	
FC05	0x0222	546	HumidityFanON	Humidity Fan ON	—	0xFF00	COIL	W	
FC05	0x0222	546	HumidityFanOFF	Humidity Fan OFF	—	0x0000	COIL	W	
FC16	0x0360	864	SetPoint	Write Setpoint	°C	×0.1 (write val×10)	INT16	W	*Write 1 register. Example: -20.0°C → write -200 (0xFF38)*